

Assignment 2 - Amazon Products Data Analysis

Task 1 — Load & Inspect the Dataset

1. Load the dataset into your notebook.
2. Display first and last 5 rows.
3. Print `.shape` and `.info()`.
4. Show list of unique product categories.
5. Identify columns that require cleaning or type conversion.

Task 2 — Clean the Dataset

1. Identify missing values using `.isnull().sum()`.
2. Fill missing numeric values (`discount_price`, `rating`, etc.) using mean or median.
3. Fill missing categorical values (`brand`, `category`) with "Unknown".
4. Remove all duplicate rows.
5. Convert numeric columns to their correct data types.
6. Reset index after cleaning.

Task 3 — Data Analysis

1. Identify the **top 5 most expensive products** (`actual_price`).
2. Find the **brand with the highest number of products**.
3. Compute the **average discount percentage** for each brand.
4. Identify **products with rating ≥ 4.5 and rating_count ≥ 2000** .
5. Compute **category-wise average price**.

Task 4 — *Programming Task*: Feature Engineering + Logical Filters

Task 4A — Create New Columns

Write code to create:

1. `discount_percent` = $((\text{actual_price} - \text{discount_price}) / \text{actual_price}) * 100$

`price_category` based on:

2.
 - `actual_price < 500` → "Budget"
 - `500 ≤ price < 2000` → "Midrange"
 - `price ≥ 2000` → "Premium"
 3. `popularity_score = rating * log(rating_count + 1)`
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Task 4B — Filtering Using New Columns

Using the newly created columns:

1. Show **top 10 highest popularity_score** products.
2. Show all **Premium** category items with **discount_percent > 40%**.
3. Show all products whose title contains "Bluetooth" (string filter).

Show products where:

4.
 - `rating ≥ 4.0`
 - `discount_percent` between 20% to 50%
 - `popularity_score` in the top 30% percentile
(Use `.between()` and `.quantile()`)
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Task 4C — Grouping with New Columns

1. Find the **average discount_percent** for each brand.
 2. Find the **average popularity_score** for each `price_category`.
 3. Identify which **category** offers the **highest average discount**.
 4. Calculate **total revenue potential** per brand using:
 5. `revenue_potential = actual_price * rating_count`
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Task 5 — Data Visualization

Use the **specific plot type** mentioned:

1. Price distribution → **Histogram + KDE**

2. Relationship between price and rating → **Scatter Plot**
3. Category-wise average rating → **Bar Plot**
4. Brand product count → **Count Plot**
5. Numeric feature correlations → **Heatmap**